

Handoff #6: the in-loop results are in, and the single call wins

Everything since handoff #5 (commit a080a1f): the Kev-4B cascade row finished, the live Jev → GPT-5.6 cascade, U4 (all five in-loop arms with GPT-5.6 as G and as the user simulator) and U5 (the paper figures, embedded below).

TL;DR

- Live Jev → GPT-5.6 cascade ($t = 0.65$): strict 0.849 [0.828, 0.869], 16.7% escalated, $8.2e-4$ \$/decision. +3.5 pts over Jev, -3.6 under GPT-5.6 (both significant), at 21% of GPT-5.6's cost. It matches the simulation (0.850).
- U4 done (50 tasks × 5 trials × 5 arms, \$74). Reward: B G-decides 0.796, C G-alone 0.792, A Jev 0.744, D GPT-OSS 0.692, E Kev-4B 0.045.
- The single call wins. C, where G decides inside its own call, is the cheapest arm (\$0.033 per success) and ties B. A separate decision step doesn't pay off against it with a frontier G.
- Among separate deciders, Jev is by far the cheapest: $A - B = -5.2$ pts reward (borderline), but -\$0.105 and -24 s per episode. Jev's decision is 5% of the agent's \$; GPT-5.6's is 75%.
- U5 done: figures 1-14 and the tables. Open question for the advisor: how to frame the paper, and whether to build router and cache-guard (page 7).

Your first 5 minutes

```
git checkout main && git pull # everything is merged into main (fast-forward)
less results/inloop/airline-frontier-gpt-5.6-sol/report/summary.md # U4: arms + paired diffs
ls results/replay/airline/paper_figures results/inloop/*/paper_figures # all figures
less DECISIONS.md # new rows dated 2026-10-02/03/04
```

1 · Where we are

Experimental arms (agent control, tau2 airline)

Arm	What	Status
0	each decider -> none (decision quality + cost)	DONE — 9 rows incl. GPT-5.6, the live Jev cascade and the finished Kev-4B cascade
A	Jev (API) -> G	DONE — frontier G: 0.744
B	G choice call -> G	DONE — frontier G: 0.796 (local G: 0.308)
C	G, one call (production default)	DONE — frontier G: 0.792 (local G: 0.196)
D	small LLM (GPT-OSS-20B) -> G	DONE — frontier G: 0.692
E	Kev-4B (local GPU) -> G	DONE — frontier G: 0.045 (local G: 0.048)

G = executor and user simulator = openrouter/openai/gpt-5.6-sol; local-G runs (Qwen3-8B) are from handoff #4.

PLAN.md checklist

- Done: G1-G4, U0-U3, and now U4 (frontier-G in-loop) and U5 (figures and tables).
- Not started: the paper's prose; router and cache-guard decision points; the floor classifier.
- Housekeeping: jev-cascade-gpt56-pilot is merged into main.

Spend this round (OpenRouter)

Run	Job(s)	Cost
Kev-4B cascade, last 4,005 calls	11865682	\$0.63
Live Jev -> GPT-5.6 cascade, 5,380 calls	11873642	\$4.39
U4 pilot, 20 episodes	11873846	\$0.89
U4 full, 1,230 episodes	11878151-53, 11878918, 11878944-46	\$73.18
Total		\$79.09

Balance left ≈ \$9. The 444 episodes refused by the key limit cost \$0 (page 8).

2 · Arm 0, final

Decider	Strict [95% CI]	Mean latency	\$ / decision	ECE	Failures
GPT-5.6-sol (ceiling) †	0.885 [0.868, 0.901]	4.90 s	3.83e-3	0.103	0.1%
Jev -> GPT-5.6, †=0.65 (live)	0.849 [0.828, 0.869]	0.63 s	8.16e-4	0.063	0.0%
Jev	0.814 [0.791, 0.836]	0.18 s	1.73e-4	0.029	0.0%
GPT-OSS-20B	0.670 [0.646, 0.693]	1.99 s	3.11e-4	0.286	10.9%
Kev-4B -> GPT-OSS, †=0.9	0.647 [0.623, 0.672]	3.94 s	2.11e-4	0.311	11.3%
Kev-9B / Qwen3-8B / Kev-4B / Kev-0.8B	0.401 / 0.396 / 0.321 / 0.273	0.24-0.72 s	local	—	0-3.3%

† Graded against its own reference actions, so an upper bound. The Kev-4B cascade is now complete; it escalated every successful call in its final job.

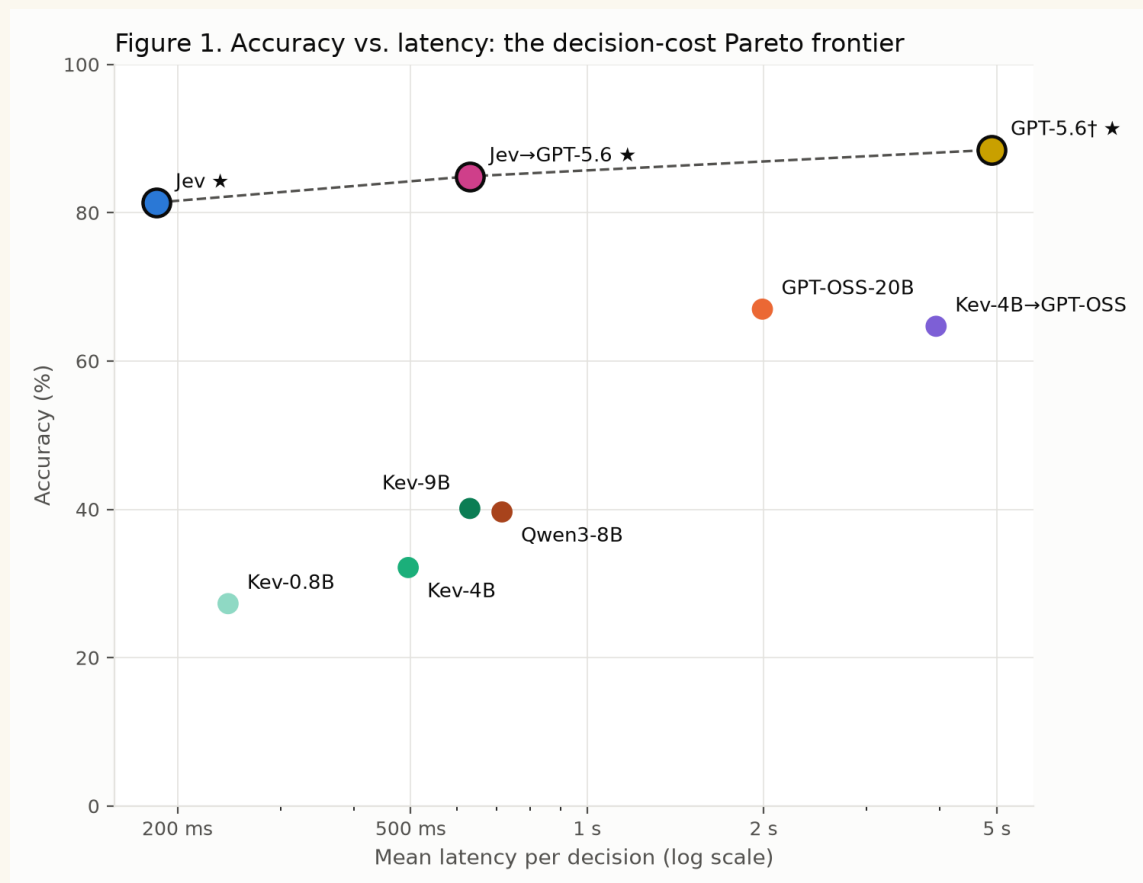


Figure 1 now uses mean latency: on the median, the cascade (5 of 6 calls never escalate) showed as faster than Jev on run-to-run noise. The frontier is Jev -> cascade -> GPT-5.6.

3 · The live Jev -> GPT-5.6 cascade

$t = 0.65$	Accuracy	Sent to GPT-5.6	\$ / decision	Mean latency
Simulated (from the standalone runs)	0.850 (0.854 held-out)	16.7%	8.36e-4	1.09 s
Live run (job 11873642)	0.849	16.7%	8.16e-4	0.63 s

Paired over 1,076 states: cascade - Jev +3.5 pts [+2.2, +4.9]; cascade - GPT-5.6 -3.6 pts [-5.0, -2.2]. Accuracy and cost match the simulation; the live run was faster because its escalated GPT-5.6 calls ran faster than in the standalone replay.

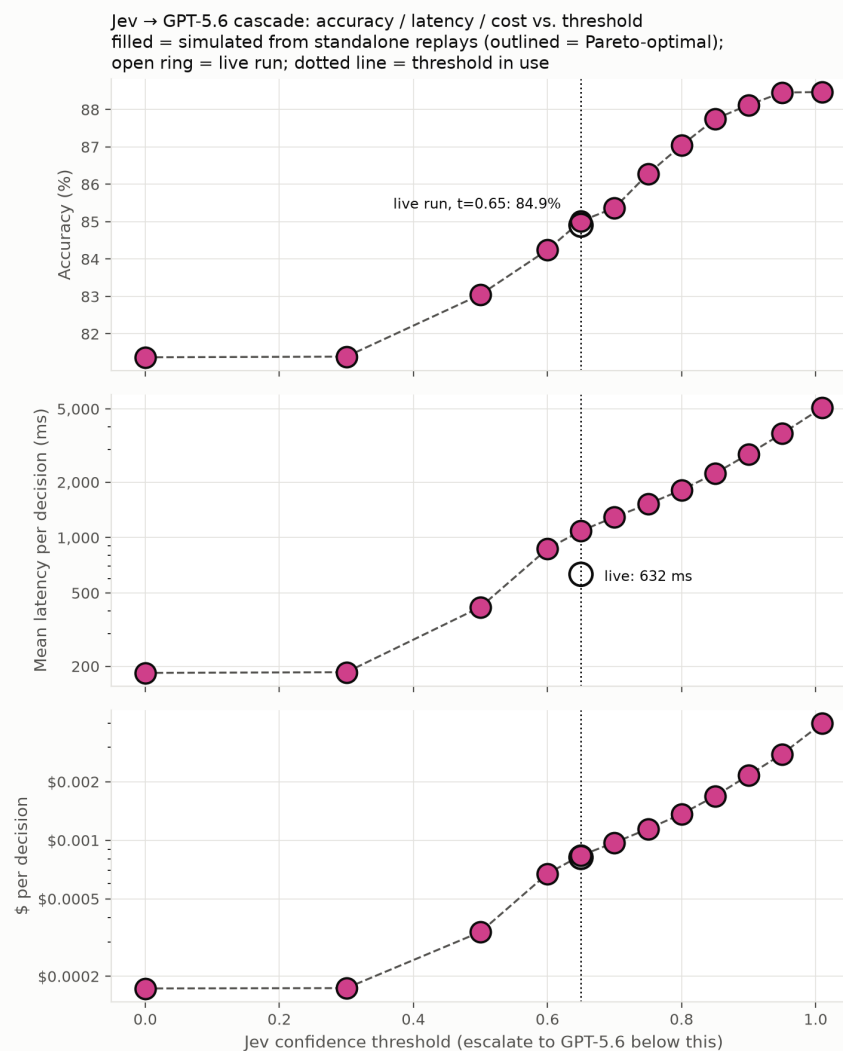


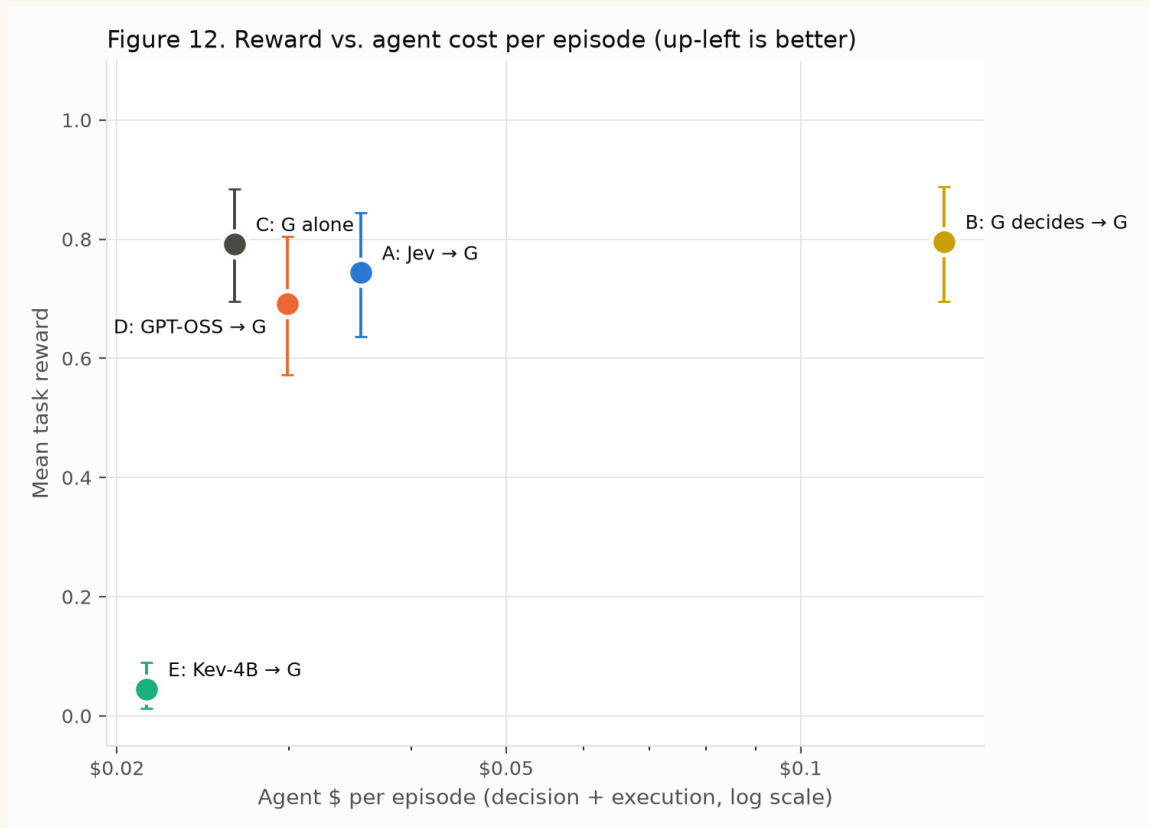
Figure 10. The dotted line is the threshold in use (0.65, chosen on held-out tasks). The simulation now includes Jev's own \$ per call.

4 · U4: the five arms in the loop

Airline, 50 tasks × 5 trials per arm, same user-simulator seed per (task, trial) across arms. G and the user simulator are GPT-5.6-sol.

Arm	Reward [95% CI]	pass ⁵	Agent \$ / episode	Agent \$ / success	Decision share of \$	Wall s / ep
B: G decides → G	0.796 [0.696, 0.888]	0.660	\$0.140	\$0.176	74.5%	65.6
C: G alone (1 call)	0.792 [0.696, 0.884]	0.640	\$0.026	\$0.033	0%	37.3
A: Jev → G	0.744 [0.636, 0.844]	0.600	\$0.036	\$0.048	5.3%	41.4
D: GPT-OSS-20B → G	0.692 [0.572, 0.804]	0.540	\$0.030	\$0.043	4.9%	59.8
E: Kev-4B → G	0.045 [0.012, 0.089]	—	\$0.022	\$0.476	0% (local)	31.5

Agent \$ = decision + execution; the user simulator's \$ is reported apart. Arm E: 244 episodes plus 6 where G returned an empty message (counted as errors).

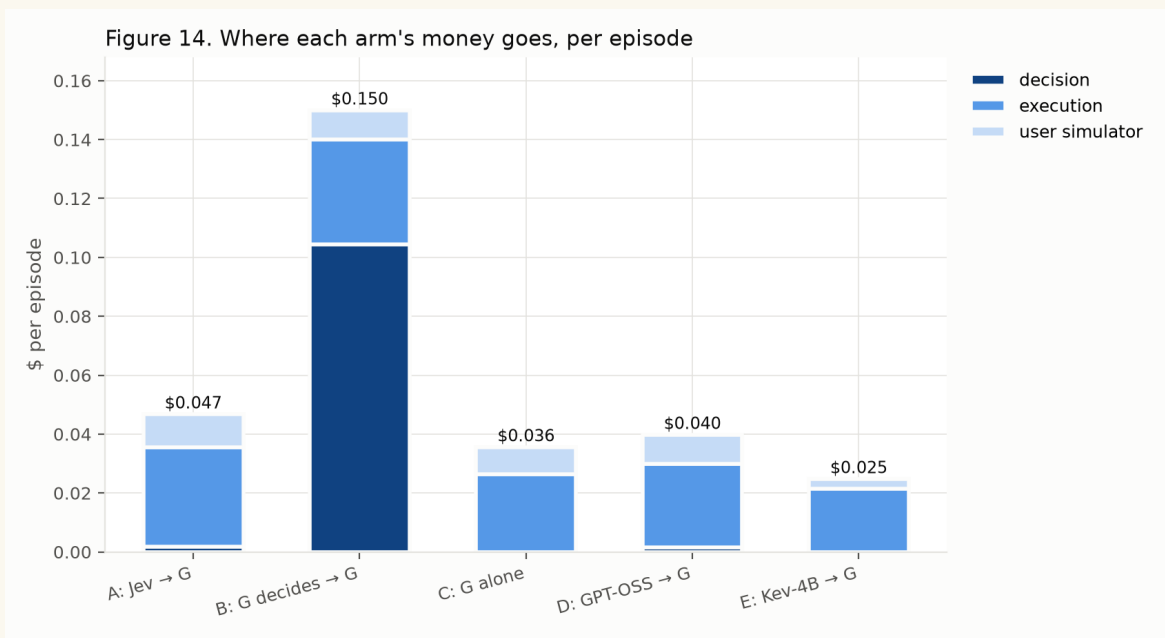


5 · Paired arm differences

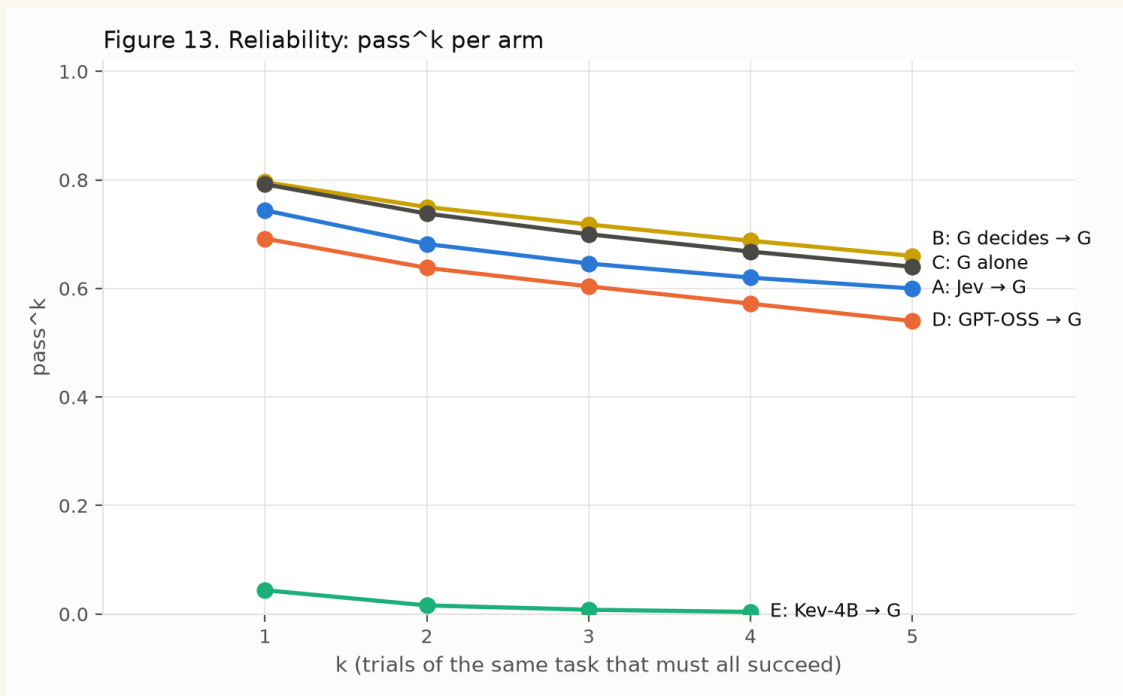
Comparison	Reward diff [95% CI]	Agent \$ / episode diff	Wall s / episode diff
A – B: Jev vs G deciding (same executor)	–0.052 [–0.108, –0.004] *	–\$0.105 *	–24.2 s *
A – C: Jev + G vs the single call	–0.048 [–0.108, +0.008]	+\$0.009 *	+4.1 s *
B – C: G deciding separately vs single call	+0.004 [–0.028, +0.040]	+\$0.114 *	+28.3 s *
D – A: GPT-OSS vs Jev deciding	–0.052 [–0.124, +0.020]	–\$0.006 *	+18.4 s *
E – A: Kev-4B vs Jev deciding	–0.701 [–0.801, –0.595] *	–\$0.014 *	–9.9 s *

Per-task mean difference over the same (task, trial) pairs, 95% bootstrap CI over 50 tasks; * = CI excludes 0. A – B's CI only just excludes 0 (it touched 0 on a partial-data run), so call it borderline.

Reading it honestly: with a frontier G, splitting decide from execute doesn't beat letting G decide in its own call. C ties B on reward at a fifth of the cost, and is cheaper than A, because the split adds a call every turn. Where Jev wins is against a separate frontier decision step: A – B saves \$0.105 and 24 s per episode for about 5 pts of reward, the same gap arm 0 shows (Jev 0.814 vs GPT–5.6 0.885 per decision). The arm-0 cascade closes half that gap, but it hasn't been run in the loop.



6 · Reliability, and the question for the advisor



pass^k keeps the same ranking at every k : $B \approx C > A > D$, with Kev-4B near 0.

Decide before building more

The project's premise was that a cheap decider saves money against the frontier model doing everything. U4 says that holds against a separate frontier decision step (A vs B), but not against the frontier model's own single call (C). Three ways forward, best discussed with the advisor:

- Frame the paper around this result (agent control only): the numbers and figures exist; what's left is prose.
- Test the in-loop cascade first (~\$15-20 plus a little code): Jev decides, and GPT-5.6 decides below $t = 0.65$. If it closes A's 5-pt gap at near C's cost, Jev has a clear role.
- Keep the full scope: build router and cache-guard (no harness or dataset yet; ~\$50-75 API plus 1-2 weeks each).

7 · What's next

- Advisor meeting: paper framing and scope (page 7).
- Paper prose from figures 1-14 and the tables. State plainly: Kev is not Jev; GPT-5.6's arm-0 accuracy is an upper bound; airline is the only headline domain.
- Credit: ~\$9 left. Load ~\$25 only for the in-loop cascade; ~\$75 for router + cache-guard once built.

Gotchas we hit this round

- OpenRouter keys have their own spend limit, separate from the account balance (Settings > API Keys). Its refusal ("Key limit exceeded", HTTP 403) wasn't recognized, so 444 episodes failed instantly. Fixed in `dwg/errors.py`: it now stops paid calls and is retried on resume. Raise the key limit before big runs.
- davisjam's normal-QoS GPU cap is 3× A100-80GB and 0 for every other card; standby doesn't count against it. A job moved to standby after submitting must also drop the group GRES: `scontrol update JobId=<id> TresPerNode=gres/gpu:1`.
- All gilbreth-k nodes failed every job launch after a reboot on 2026-10-03 and recovered on 10-04. `sacct -D` shows the node of each failed launch.
- `results/inloop/` wasn't in git (`.gitignore` only whitelisted states and replay). Now it is.
- Arm B costs more than its pilot said (\$0.15 vs \$0.10 per episode): a 4-task pilot underestimates. Give per-arm caps headroom.

Changed files this handoff

```
scripts/jev_cascade_replay.slurm, scripts/inloop_frontier.slurm # new: live cascade, frontier-G
in-loop
scripts/make_inloop_figures.py # new: figs 11-14
scripts/make_paper_figures.py, make_cascade_figure.py, post_experiment_analysis.py # GPT-5.6,
Jev cascade
scripts/analyze_inloop.py # paired arm differences
src/dwg/errors.py, tests/test_analyze.py # key limit = billing error
results/inloop/airline-frontier-gpt-5.6-sol/, results/replay/airline/ # data + figures
.gitignore, DECISIONS.md, PLAN.md, scripts/make_handoff6_pdf.py
```